

The real Hilbert endpoint in Raynaud's $\mathcal{BL}_1L_q$ axiomatizability problem

Proof draft for independent verification

June 18, 2026

Abstract

Raynaud asked whether the Banach-space reduct of the mixed-norm Banach-lattice class $\mathcal{BL}_1L_q$ is axiomatizable for $1 < q \leq 2$. This note proves the remaining real Hilbert endpoint $q = 2$.

The main new device is a canonical radial truncation operation β on L_1 -direct integrals of real Hilbert spaces. Although the lattice structure is not determined by the Banach norm at $q = 2$, the operation β is: it admits an intrinsic description using the Boolean algebra of L -projections and is therefore preserved by every surjective linear isometry. It also commutes with ultraproducts. A Keisler–Shelah common-ultrapower/equalizer argument then reduces ultraroot closure to a geometric statement about closed β -invariant subspaces. Such a subspace is shown to be an ℓ_1 -sum of Hilbert-valued L_1 -spaces by extracting a scalar sub- σ -algebra and applying Raynaud's module representation lemma. Consequently the real Banach-space class underlying $\mathcal{BL}_1L_2$ is closed under ultraproducts and ultraroots, hence is axiomatizable.

A targeted literature search through June 18, 2026 did not locate this endpoint argument in print. The note is therefore presented as a complete proof draft for independent expert verification before citation as an established result.

Packet status

This packet is a candidate full solution of the real endpoint $q = 2$ of Raynaud's Question 6.17. It should be read as a proof draft for independent expert verification, not as a human-verified result. The active claim is positive: the Banach-space reduct of $\mathcal{BL}_1L_2$ is axiomatizable.

An older packet in this run attempted a negative result for fiber-dimension-restricted variants using a stable-variable ultrapower route. That older route is not used here and is stored only as history: mutual finite representability gives ultrapower embeddings, not the surjective ultrapower identification that the old argument needed.

Source problem statement.

space, see [40]. □

The main question remaining open in the context of $\mathcal{BL}_p L_q$ -spaces is:

Question 6.17. Is the class $\mathcal{BL}_1 L_q$ axiomatizable in the language of Banach spaces, for $1 < q \leq 2$?

REFERENCES

1 Statement of the result

For $1 \leq p, q < \infty$, let $\mathcal{BL}_p L_q$ denote the class of Banach lattices lattice-isometric to bands in mixed-norm lattices $L_p(\Omega; L_q(S))$. If $\mathcal{A}$ is a class of Banach lattices, write $\mathcal{A}^B$ for the class of their underlying Banach spaces, up to surjective linear isometry. Raynaud proved many Banach-space axiomatizability results for these classes and left the following range open [7, Question 6.17]:

$$(\mathcal{BL}_1 L_q)^B, \quad 1 < q \leq 2.$$

The purpose of this note is to settle the real endpoint.

Theorem 1.1 (Main theorem). *The class of real Banach spaces linearly isometric to members of $\mathcal{BL}_1 L_2$ is axiomatizable in the language of real Banach spaces. Equivalently,*

$$\boxed{(\mathcal{BL}_1 L_2)^B \text{ is closed under ultraroots.}}$$

Closure under ultraproducts was already known from the Banach-lattice theory of $\mathcal{BL}_p L_q$ [3, 7]. Thus only ultraroot closure needs proof.

The argument does not reconstruct the coordinate lattice in a Hilbert fibre. That would be impossible: arbitrary orthogonal operators preserve the norm and need not preserve lattice disjointness. Instead, it recovers a weaker radial operation which is exactly strong enough to recognize the underlying Banach-space class.

2 The endpoint class and random norms

Let H be a real Hilbert space and let (Ω, Σ, μ) be a localizable measure space. For $x \in L_1(\Omega; H)$ write

$$N(x)(\omega) = \|x(\omega)\|_H$$

for its random norm. We use the same notation in bands and direct integrals.

Define $\mathcal{C}_1^{\mathbb{R}}$ to be the class of real Banach spaces isometric to spaces of the form

$$\left(\bigoplus_{\gamma \in \Gamma} L_1(\Omega_\gamma, \mu_\gamma; H_\gamma) \right)_1, \tag{1}$$

where the H_γ are real Hilbert spaces and the bases are localizable.

Proposition 2.1. *One has*

$$\mathcal{C}_1^{\mathbb{R}} = (\mathcal{BL}_1 L_2)^B.$$

Moreover every member of $\mathcal{C}_1^{\mathbb{R}}$ can be represented as a band, hence as a closed $L_\infty(\Sigma)$ -submodule, of some $L_1(\Omega; H)$ with a fixed real Hilbert space H .

Proof. Using the standard localizable realization of the base, a band in $L_1(\Omega; L_2(S))$ is a closed $L_\infty(\Sigma)$ -submodule of $L_1(\Omega; H)$, where $H = L_2(S)$, and its random norms are Σ -measurable. Raynaud's module representation lemma [6, Lemma 4.4] (with $p = 1$ and $\mathcal{F} = \Sigma$) gives a random-norm-preserving isometry onto an ℓ_1 -sum of spaces $L_1(\Omega_\gamma; H_\gamma)$.

Conversely, choose an orthonormal basis in every H_γ , identify it with a coordinate band of one sufficiently large Hilbert lattice $\ell_2(I)$, and take the disjoint union of the base spaces. The space in (1) is then a band in $L_1(\Omega; \ell_2(I))$. $\square$

We shall also use the mixed-norm ultraproduct theorem of Lévy–Raynaud and Henson–Raynaud [5, 3, 4]. In the form needed here it says the following.

Proposition 2.2 (Random norms in ultraproducts). *Let $(Z_i)_{i \in I}$ be members of $\mathcal{C}_1^{\mathbb{R}}$, each equipped with one of its $\mathcal{BL}_1 L_2$ lattice representations and random norm N_i . For an ultrafilter $\mathcal{U}$ on I , the Banach ultraproduct $Z = \prod_{i \in \mathcal{U}} Z_i$ admits a $\mathcal{BL}_1 L_2$ representation with random norm N satisfying*

$$N([x_i]_{\mathcal{U}}) = [N_i(x_i)]_{\mathcal{U}}.$$

In particular, $\mathcal{C}_1^{\mathbb{R}}$ is closed under Banach ultraproducts.

3 A canonical radial operation

For a direct-integral representation $Z = L_1(\Omega; H_\omega)$, define

$$\beta_Z(x, y)(\omega) = \begin{cases} \frac{\min\{N(x)(\omega), N(y)(\omega)\}}{N(y)(\omega)} y(\omega), & N(y)(\omega) > 0, \\ 0, & N(y)(\omega) = 0. \end{cases} \quad (2)$$

Thus $\beta_Z(x, y)(\omega)$ is the radial projection of $y(\omega)$ onto the Hilbert ball of radius $N(x)(\omega)$. We next prove that this operation is intrinsic to the Banach norm.

Recall that an L -projection on a Banach space Z is a bounded projection P such that

$$\|z\| = \|Pz\| + \|(I - P)z\| \quad (z \in Z).$$

Lemma 3.1 (The base is intrinsic). *Let $Z = L_1(\Omega; H_\omega)$ be an L_1 -direct integral of nonzero real Hilbert spaces. Its L -projections are exactly the base multipliers*

$$P_A z = \mathbf{1}_A z \quad (A \in \Sigma),$$

modulo null sets.

Proof. Every P_A is plainly an L -projection. Conversely, let P be an L -projection, $R = \text{Ran } P$, and $K = \ker P$. For $r \in R$ and $k \in K$,

$$\|r + k\| = \|r\| + \|k\|, \quad \|r - k\| = \|r\| + \|k\|.$$

Both pointwise triangle defects are nonnegative and have integral zero. Hence both triangle equalities hold almost everywhere in every Hilbert fibre. Simultaneous equality for $r(\omega) + k(\omega)$ and $r(\omega) - k(\omega)$ forces $r(\omega) = 0$ or $k(\omega) = 0$.

In the complete measure algebra, let A be the supremum of the supports of all elements of R . Every element of R is supported in A , and the preceding base-disjointness forces every element of K to be supported in A^c . If $z = Pz + (I - P)z$, it follows that $Pz = \mathbf{1}_A z$. $\square$

For L -projections Q, P , write $Q \leq P$ when $\text{Ran } Q \subseteq \text{Ran } P$. For $x, y \in Z$ and an L -projection P , put

$$m_{x,y}(P) = \inf_{Q \leq P} (\|Qx\| + \|(P - Q)y\|). \quad (3)$$

By Lemma 3.1, if $P = P_A$, then

$$m_{x,y}(P_A) = \int_A \min\{N(x), N(y)\} d\mu. \quad (4)$$

Indeed one minimizes pointwise by taking $Q = P_{A \cap \{N(x) \leq N(y)\}}$.

Lemma 3.2 (Intrinsic characterization). *For $x, y \in Z$, the vector $z = \beta_Z(x, y)$ is the unique vector satisfying*

$$\|Pz\| = m_{x,y}(P) \quad \text{for every } L\text{-projection } P, \quad (5)$$

$$\|y\| = \|z\| + \|y - z\|. \quad (6)$$

Consequently every surjective linear isometry $T : Z \rightarrow Z'$ between members of $\mathcal{C}_1^{\mathbb{R}}$ satisfies

$$T\beta_Z(x, y) = \beta_{Z'}(Tx, Ty).$$

In particular, β_Z is independent of the chosen direct-integral or band representation of Z .

Proof. Formula (2) gives $N(z) = \min\{N(x), N(y)\}$, so (5) follows from (4). Also $z(\omega)$ is a scalar multiple of $y(\omega)$ with coefficient in $[0, 1]$, and hence (6) holds.

Conversely, (5) and (4), applied to every base set, imply

$$N(z) = \min\{N(x), N(y)\} \quad \text{a.e.}$$

The pointwise inequality $N(y) \leq N(z) + N(y - z)$ has equal integrals by (6); it is therefore an equality almost everywhere. Equality in the triangle inequality of a strictly convex Hilbert space implies $z(\omega) = t(\omega)y(\omega)$ with $0 \leq t(\omega) \leq 1$. The random-norm identity forces

$$t(\omega) = \frac{\min\{N(x)(\omega), N(y)(\omega)\}}{N(y)(\omega)}$$

on $\{N(y) > 0\}$, proving uniqueness.

A surjective isometry conjugates the Boolean algebras of L -projections, preserves their order, and preserves every norm occurring in (3)–(6). It therefore sends the unique solution for (x, y) to the unique solution for (Tx, Ty) . $\square$

Lemma 3.3 (Uniform continuity). *For every $Z \in \mathcal{C}_1^{\mathbb{R}}$,*

$$\|\beta_Z(x, y) - \beta_Z(x', y')\| \leq \|x - x'\| + \|y - y'\|.$$

Proof. For a Hilbert space, let Q_r be the metric projection onto the closed ball of radius r . Metric projections onto closed convex sets are nonexpansive, and directly $\|Q_r v - Q_s v\| \leq |r - s|$. Hence pointwise

$$\|Q_{\|u\|} v - Q_{\|u'\|} v'\| \leq \|u - u'\| + \|v - v'\|.$$

Integrating gives the assertion. $\square$

Lemma 3.4 (Ultraproduct compatibility). *If $Z = \prod_{i \in \mathcal{U}} Z_i$ is a Banach ultraproduct of members of $\mathcal{C}_1^{\mathbb{R}}$, then*

$$\beta_Z([x_i]_{\mathcal{U}}, [y_i]_{\mathcal{U}}) = [\beta_{Z_i}(x_i, y_i)]_{\mathcal{U}}.$$

Proof. Choose $\mathcal{BL}_1 L_2$ lattice representations and use Proposition 2.2. Put $x = [x_i]_{\mathcal{U}}$, $y = [y_i]_{\mathcal{U}}$, and $z = [\beta_{Z_i}(x_i, y_i)]_{\mathcal{U}}$. The coordinate identities give

$$\begin{aligned} N(z) &= N(x) \wedge N(y), \\ N(y) &= N(z) + N(y - z). \end{aligned}$$

In the resulting $L_1(L_2)$ representation, the second equality says that z and $y - z$ have equality in the Hilbert triangle inequality in almost every fibre. Thus $z = ty$ for a scalar $0 \leq t \leq 1$, and the first equality identifies t with the radial coefficient in (2). Therefore $z = \beta_Z(x, y)$. Representation independence follows from Lemma 3.2. $\square$

4 A common-ultrapower/equalizer transfer lemma

We isolate the model-theoretic mechanism. No new operation is added to the Banach-space language; the operation below is used only as an external canonical invariant.

Theorem 4.1 (Canonical-operation transfer). *Let $\mathcal{C}$ be an isometry-invariant class of Banach spaces closed under ultraproducts. Suppose that every $Z \in \mathcal{C}$ carries a binary operation β_Z such that:*

- (a) *the operations are uniformly continuous with a common modulus;*
- (b) *every surjective linear isometry between members of $\mathcal{C}$ preserves β ;*
- (c) *β commutes with Banach ultraproducts and diagonal embeddings;*
- (d) *every closed linear subspace $E \subseteq Z \in \mathcal{C}$ satisfying $\beta_Z(E, E) \subseteq E$ belongs to $\mathcal{C}$.*

Then $\mathcal{C}$ is closed under ultraroots.

Proof. Suppose that $X^{\mathcal{U}}$ is linearly isometric to a member of $\mathcal{C}$. Transport that structure and write

$$Z = X^{\mathcal{U}} \in \mathcal{C}.$$

Let $i = D_X : X \rightarrow Z$ be the diagonal embedding and put

$$e_0 = D_Z : Z \longrightarrow Z^{\mathcal{U}}, \quad e_1 = S := i^{\mathcal{U}} : Z = X^{\mathcal{U}} \longrightarrow Z^{\mathcal{U}}.$$

Both e_0 and e_1 are elementary embeddings in the pure Banach-space language.

Expand that language by a constant symbol c_z for every $z \in Z$, and consider the two expanded structures

$$\mathfrak{A} = (Z^{\mathcal{U}}, (e_0 z)_{z \in Z}), \quad \mathfrak{B} = (Z^{\mathcal{U}}, (e_1 z)_{z \in Z}).$$

They are elementarily equivalent: the value of any formula involving finitely many named elements is pulled back by either elementary embedding to the same value in Z . By the metric Keisler–Shelah theorem [1, Theorem 5.7], there are an ultrafilter $\mathcal{V}$ and an isomorphism

$$T : \mathfrak{A}^{\mathcal{V}} \longrightarrow \mathfrak{B}^{\mathcal{V}}.$$

Write

$$M = (Z^{\mathcal{U}})^{\vee}, \quad j = D_{Z^{\mathcal{U}}}^{\vee} : Z^{\mathcal{U}} \longrightarrow M.$$

The underlying space M belongs to $\mathcal{C}$, and $T : M \rightarrow M$ is a surjective linear isometry. It therefore preserves β . Moreover, because the constants are part of the expanded language,

$$Tje_0z = je_1z \quad (z \in Z).$$

Using this identity, preservation of β by T , and compatibility with diagonal embeddings, we obtain for $a, b \in Z$

$$\begin{aligned} je_1\beta_Z(a, b) &= Tje_0\beta_Z(a, b) \\ &= T\beta_M(je_0a, je_0b) \\ &= \beta_M(Tje_0a, Tje_0b) \\ &= \beta_M(je_1a, je_1b) \\ &= j\beta_{Z^{\mathcal{U}}}(e_1a, e_1b). \end{aligned}$$

Since j is injective,

$$S\beta_Z(a, b) = \beta_{Z^{\mathcal{U}}}(Sa, Sb) \quad (a, b \in Z). \quad (7)$$

The diagonal copy of X in Z is the equalizer

$$i(X) = \{z \in Z : D_Z z = Sz\}. \quad (8)$$

Indeed, for $z = [x_k]_{\mathcal{U}}$ the equality $D_Z z = Sz$ is exactly

$$\lim_{k, \mathcal{U}} \|z - i(x_k)\| = 0.$$

Thus the distance from z to the closed range $i(X)$ is zero, so $z \in i(X)$; the converse is immediate.

Equations (7) and (8), together with the fact that D_Z preserves β , show that $i(X)$ is β_Z -closed. By (d), $i(X) \in \mathcal{C}$, and hence $X \in \mathcal{C}$. $\square$

Remark 4.2. The common-ultrapower step is the only model-theoretic ingredient beyond Łoś theorem. It avoids any need to choose a specially saturated or homogeneous ultrapower inside the class $\mathcal{C}$.

5 Radially closed subspaces

This section proves condition (d) of Theorem 4.1 for $\mathcal{C}_1^{\mathbb{R}}$.

Theorem 5.1 (Radial-closure representation). *Let $Z \in \mathcal{C}_1^{\mathbb{R}}$ and let $E \subseteq Z$ be a closed real linear subspace. If*

$$\beta_Z(E, E) \subseteq E,$$

then $E \in \mathcal{C}_1^{\mathbb{R}}$.

Proof. By Proposition 2.1, represent Z as a band in $L_1(\Omega, \Sigma, \mu; H)$ for a fixed real Hilbert space H . The radial operation agrees with the ambient one.

Step 1: support decomposition.

Call x, y *base-disjoint* when $N(x)N(y) = 0$ almost everywhere. Choose a maximal pairwise base-disjoint family $(e_{\alpha})_{\alpha \in A}$ of nonzero elements of E , and put $S_{\alpha} = \text{supp } N(e_{\alpha})$. If $x \in E$, then

$$\beta_Z(me_{\alpha}, x) \longrightarrow \mathbf{1}_{S_{\alpha}}x \quad (m \rightarrow \infty)$$

in L_1 , by dominated convergence. Hence every restriction $\mathbf{1}_{S_\alpha}x$ belongs to E .

For a fixed x , the set

$$A_x = \{\alpha : \|\mathbf{1}_{S_\alpha}x\| > 0\}$$

is countable, and the corresponding disjoint sum converges in norm. If $S = \bigvee_\alpha S_\alpha$ in the complete measure algebra, then for $\alpha \notin A_x$ one has $S_\alpha \wedge \text{supp } N(x) = 0$. Since finite meets distribute over arbitrary joins in a complete Boolean algebra, $N(x) = 0$ on $S \setminus \bigvee_{\alpha \in A_x} S_\alpha$. Hence

$$\mathbf{1}_S x = \sum_{\alpha \in A_x} \mathbf{1}_{S_\alpha} x \in E.$$

The residual part $\mathbf{1}_{S^c}x$ belongs to E and, if nonzero, would be base-disjoint from every e_α , contradicting maximality. Therefore

$$E = \left(\bigoplus_{\alpha \in A} E_\alpha \right)_1, \quad E_\alpha = \mathbf{1}_{S_\alpha} E. \quad (9)$$

It is enough to treat one summand. We may thus assume that E contains an element e with

$$h = N(e) > 0 \quad \text{almost everywhere on the common support of } E.$$

Step 2: normalization. Restrict to that support and set $d\nu = h d\mu$. Notice that $\nu(\Omega) = \|e\| < \infty$. Define

$$T : E \longrightarrow L_1(\Omega, \Sigma, \nu; H), \quad Tx = h^{-1}x.$$

This is a linear isometry and it intertwines radial truncation. Put $Y = T(E)$ and $u = T(e)$. Then $N(u) = 1$ almost everywhere and $\beta(Y, Y) \subseteq Y$.

Step 3: extraction of a scalar sub- σ -algebra. Define

$$A_0 = \{f \in L_1(\Omega, \Sigma, \nu) : fu \in Y\}.$$

The map $f \mapsto fu$ is an isometry, so A_0 is a closed real linear subspace of scalar L_1 and $1 \in A_0$. For $f \in A_0$,

$$\beta(fu, mu) = \min\{|f|, m\}u \longrightarrow |f|u.$$

Thus A_0 is closed under modulus and hence is a closed vector sublattice. For $y \in Y$,

$$\beta(y, mu) = \min\{N(y), m\}u \longrightarrow N(y)u,$$

so

$$N(y) \in A_0 \quad (y \in Y). \quad (10)$$

Let

$$\mathcal{F} = \{B \in \Sigma : \mathbf{1}_B \in A_0\},$$

modulo null sets. Because A_0 is a closed sublattice containing 1, $\mathcal{F}$ is a σ -algebra: complements and finite intersections follow from the lattice operations, and countable unions follow by monotone L_1 convergence. If $f \in A_0$ and $t \in \mathbb{Q}$, then

$$1 \wedge n(f - t)^+ \longrightarrow \mathbf{1}_{\{f > t\}} \quad \text{in } L_1,$$

so f is $\mathcal{F}$ -measurable. Conversely, $\mathcal{F}$ -simple functions belong to A_0 and are dense in $L_1(\mathcal{F})$. Hence

$$A_0 = L_1(\Omega, \mathcal{F}, \nu). \quad (11)$$

Step 4: module structure. If $B \in \mathcal{F}$, then $\mathbf{1}_{Bu} \in Y$, and for every $y \in Y$,

$$\beta(m\mathbf{1}_{Bu}, y) \longrightarrow \mathbf{1}_B y \quad (m \rightarrow \infty).$$

Therefore Y is closed under multiplication by indicators in $\mathcal{F}$, and then, by uniform approximation, under all multipliers in $L_\infty(\mathcal{F})$. By (10) and (11), every random norm $N(y)$ is $\mathcal{F}$ -measurable.

After completing both (Ω, Σ, ν) and $(\Omega, \mathcal{F}, \nu)$, they are finite and therefore localizable. Raynaud's module representation lemma [6, Lemma 4.4] now applies and gives pairwise almost disjoint sets $(\Omega_j)_{j \in J}$ in $\mathcal{F}$, real Hilbert spaces H_j , and a random-norm-preserving linear isometry

$$Y \cong \left(\bigoplus_{j \in J} L_1(\Omega_j, \mathcal{F}|_{\Omega_j}, \nu|_{\Omega_j}; H_j) \right)_1.$$

Thus the normalized component belongs to $\mathcal{C}_1^{\mathbb{R}}$. Undoing T and using the ℓ_1 -decomposition (9) proves $E \in \mathcal{C}_1^{\mathbb{R}}$. $\square$

6 Proof of the main theorem

Proof of Theorem 1.1. By Proposition 2.1, it is enough to prove that $\mathcal{C}_1^{\mathbb{R}}$ is axiomatizable. This class is invariant under surjective linear isometries and is closed under ultraproducts by Proposition 2.2. Lemmas 3.2, 3.3, and 3.4 show that its radial operation is canonical, uniformly continuous, and compatible with ultraproducts. Theorem 5.1 says that every closed radial-invariant subspace of a member is again in the class. The canonical-operation transfer theorem therefore implies closure under ultraroots.

A class of Banach structures is axiomatizable exactly when it is closed under isometries, ultraproducts, and ultraroots [1, Proposition 5.14]. Hence $(\mathcal{B}\mathcal{L}_1 L_2)^B$ is axiomatizable. $\square$

Corollary 6.1. *Raynaud's Question 6.17 has a positive answer at the real endpoint $q = 2$.*

7 Why the stable-variable counterargument does not apply

A proposed negative argument uses the isometric stable-variable embedding $\ell_q^n \hookrightarrow L_1$ for $1 < q \leq 2$ and concludes that L_1 and $L_1(\ell_q^n)$ have isometric ultrapowers. The first step is correct, but the conclusion is not. Heinrich's theorem characterizes finite representability by an isometric *embedding* into an ultrapower [2]; mutual finite representability yields embeddings in both directions, not a surjective isometry between ultrapowers. Hence that argument does not contradict Theorem 1.1.

Verification notes

The proof has three external inputs:

1. the mixed-norm random-norm representation of ultraproducts (Proposition 2.2);
2. the metric Keisler–Shelah theorem for the two constant expansions in the proof of Theorem 4.1;
3. Raynaud's module representation lemma [6, Lemma 4.4], used with $p = 1$.

All other ingredients, including the intrinsic characterization of β , the equalizer argument, and the extraction of $\mathcal{F}$, are proved in the note. Since the endpoint conclusion appears to be new, these three interfaces—and in particular the passage from the intrinsic radial operation to the common-ultrapower equalizer—are the natural points for independent expert verification.

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